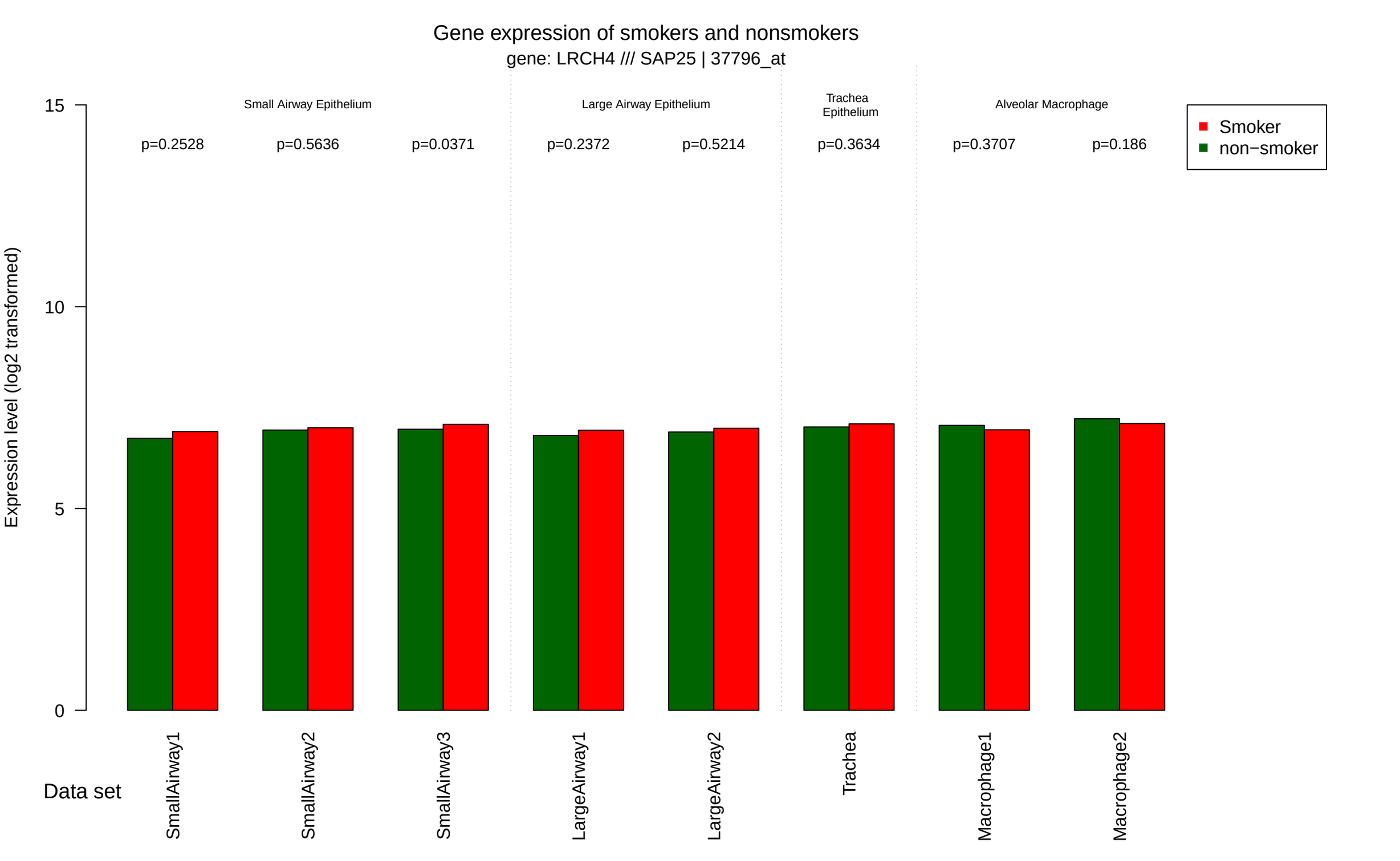
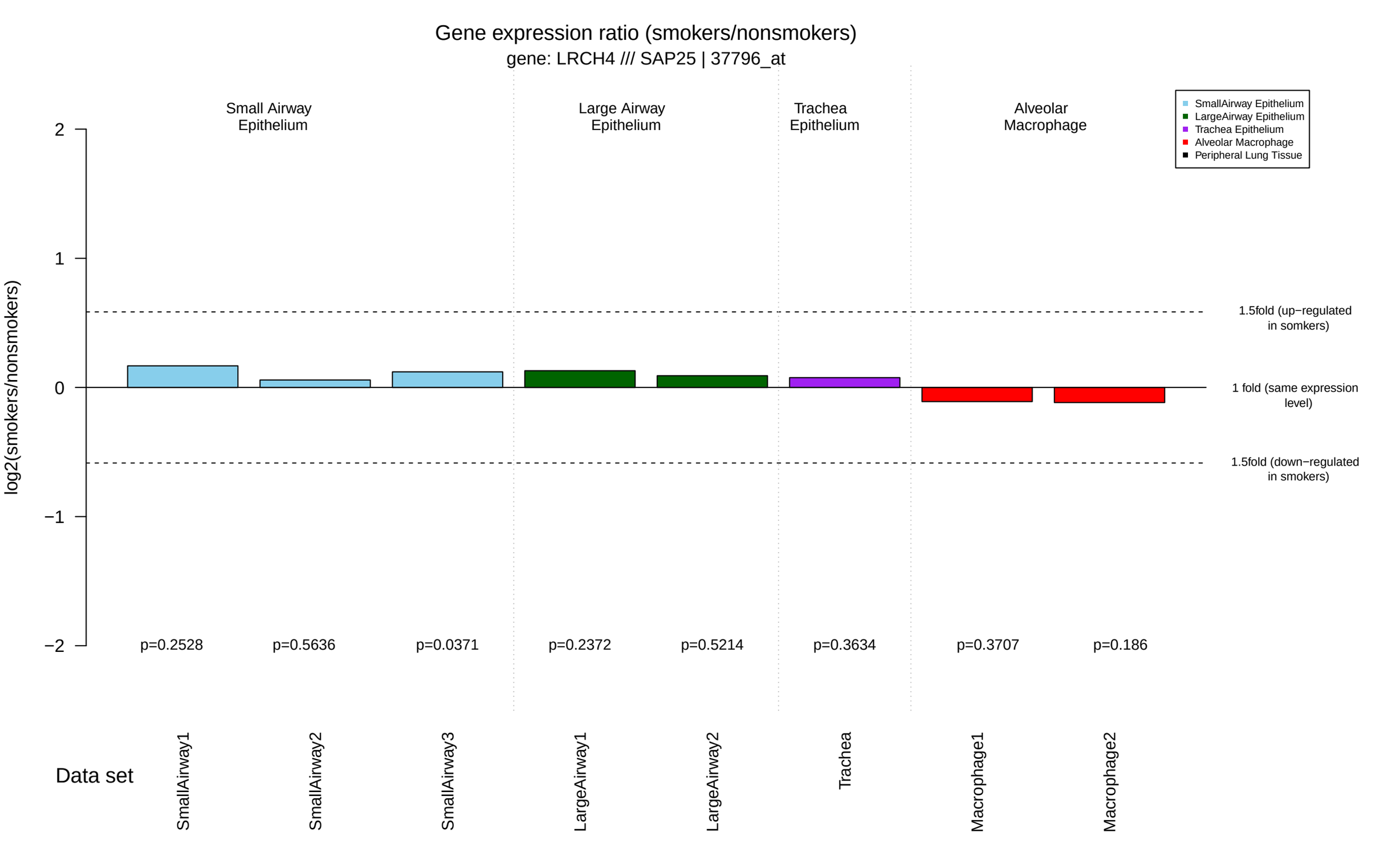
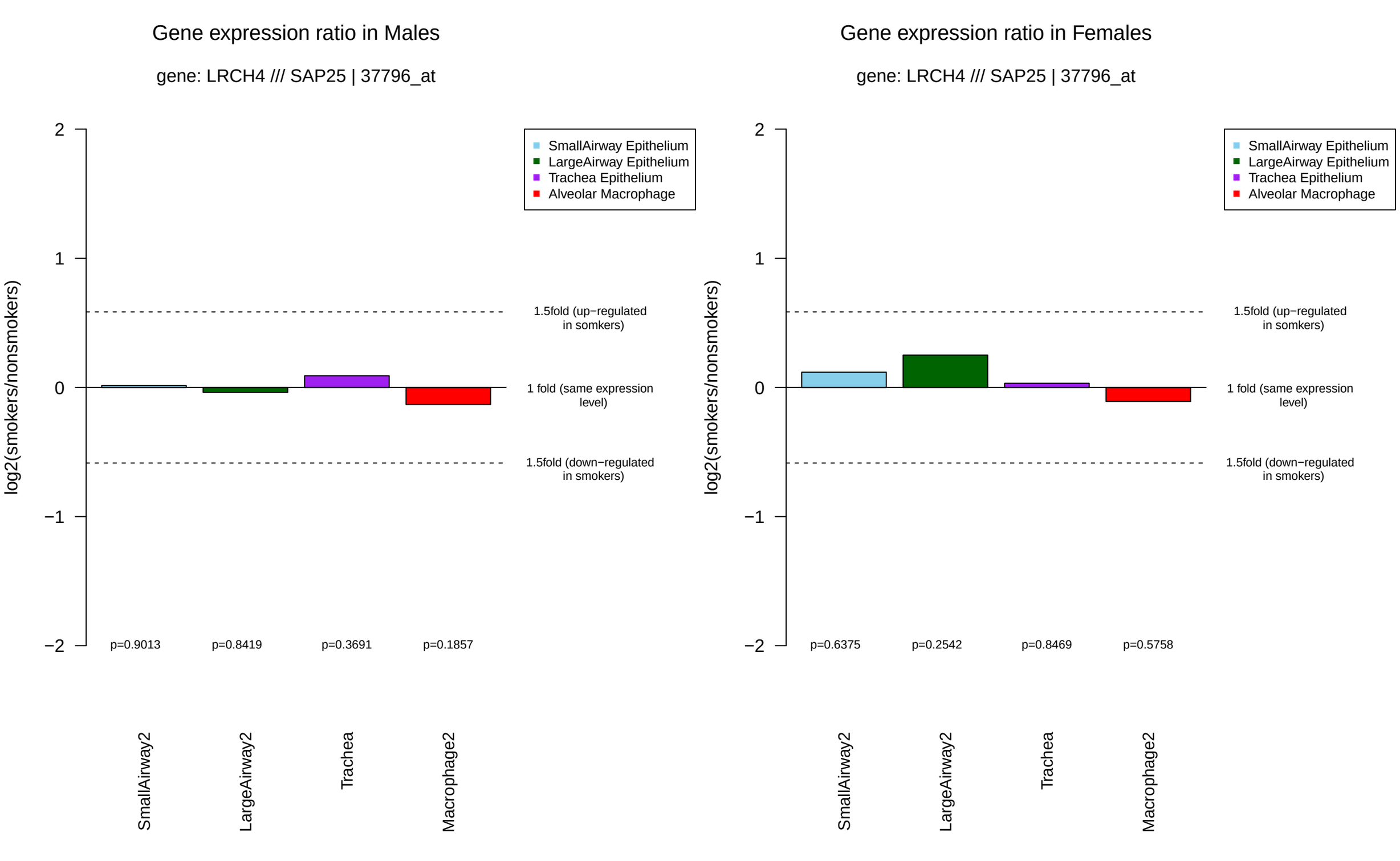
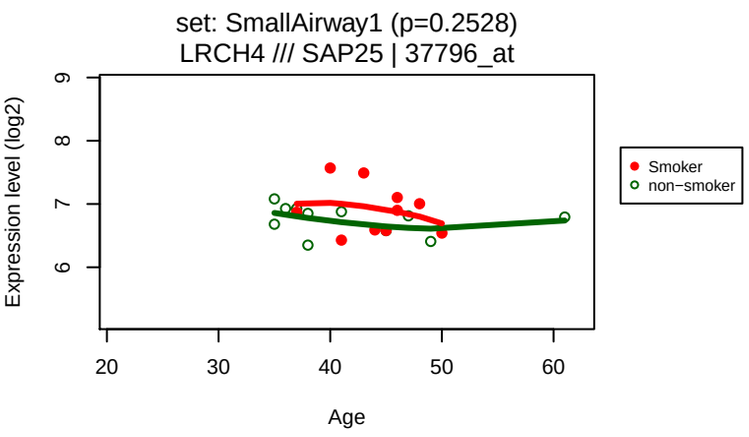
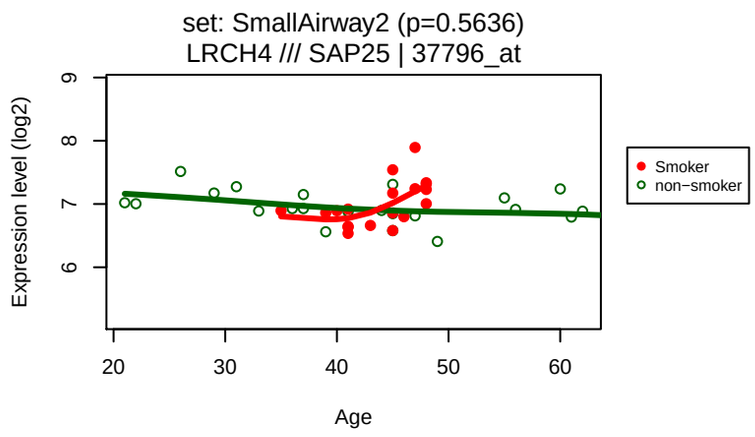
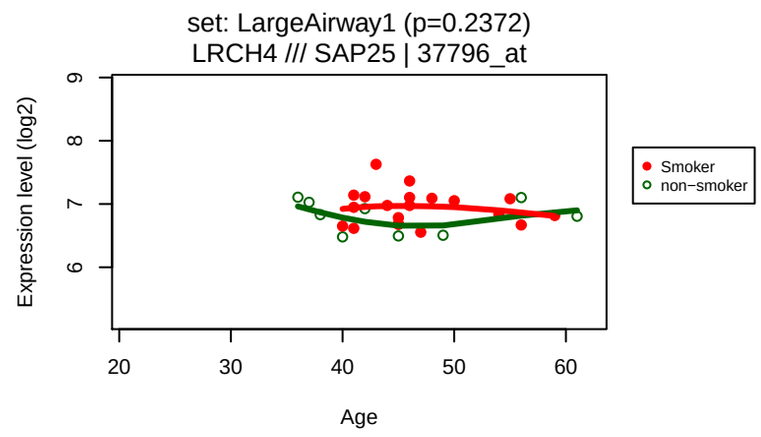

Gene expression of smokers and nonsmokers
gene: LRCH4 /// SAP25 | 37796_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2528
p=0.5636
p=0.0371
p=0.2372
p=0.5214
p=0.3634
p=0.3707
p=0.186
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

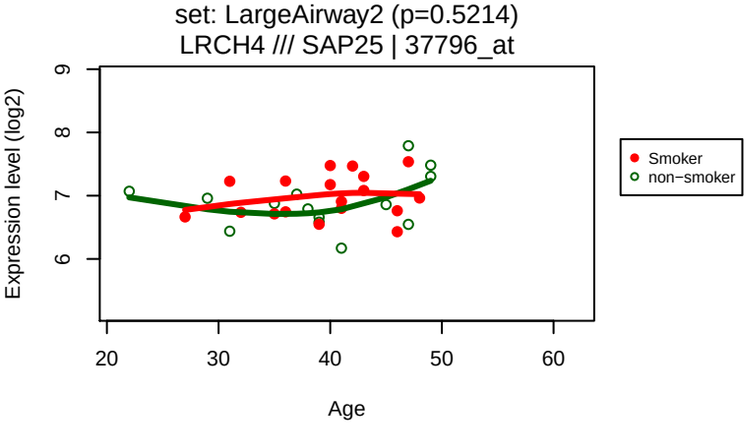

Gene expression ratio (smokers/nonsmokers)
gene: LRCH4 /// SAP25 | 37796_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2528
p=0.5636
p=0.0371
p=0.2372
p=0.5214
p=0.3634
p=0.3707
p=0.186
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

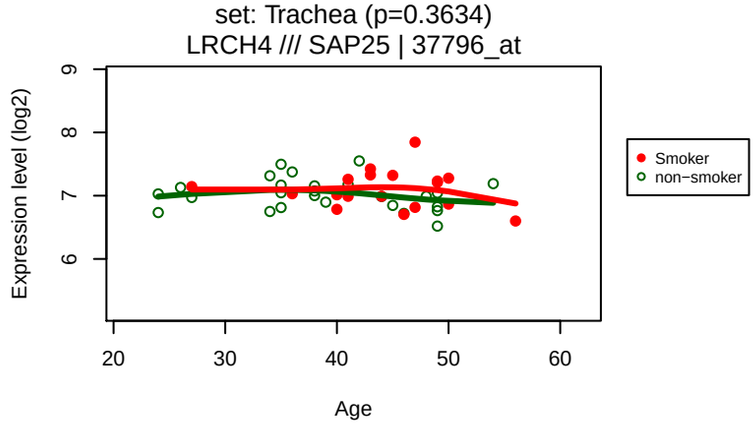

Gene expression ratio in Males
gene: LRCH4 /// SAP25 | 37796_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9013
p=0.8419
p=0.3691
p=0.1857
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LRCH4 /// SAP25 | 37796_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6375
p=0.2542
p=0.8469
p=0.5758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

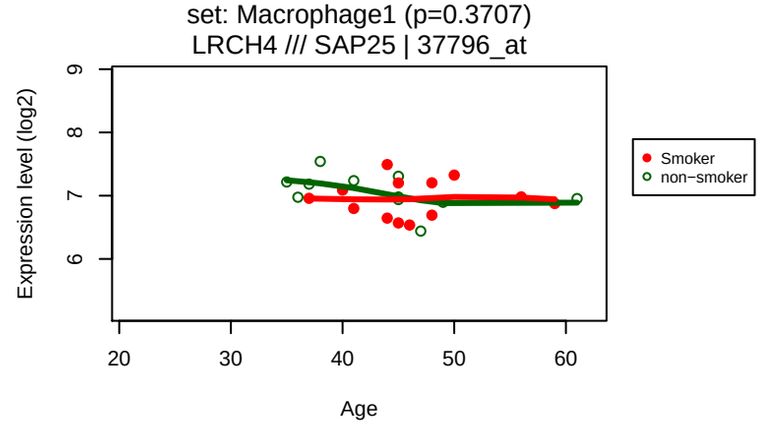

set: SmallAirway1 (p=0.2528)
LRCH4 /// SAP25 | 37796_at
Expression level (log2)
Age
Smoker
non−smoker

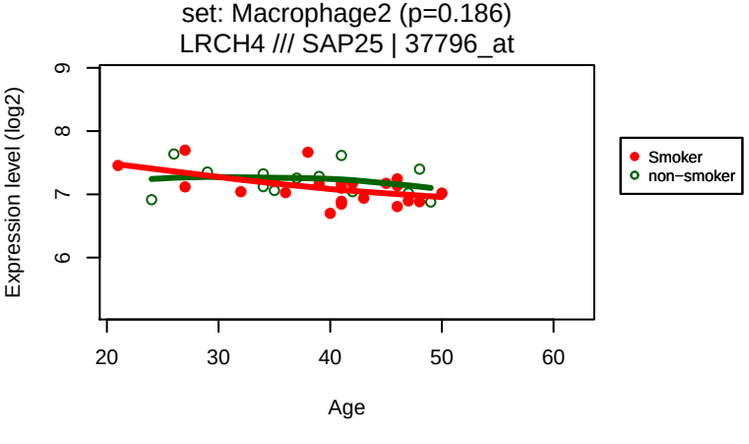

set: SmallAirway2 (p=0.5636)
LRCH4 /// SAP25 | 37796_at
Expression level (log2)
Age
Smoker
non−smoker

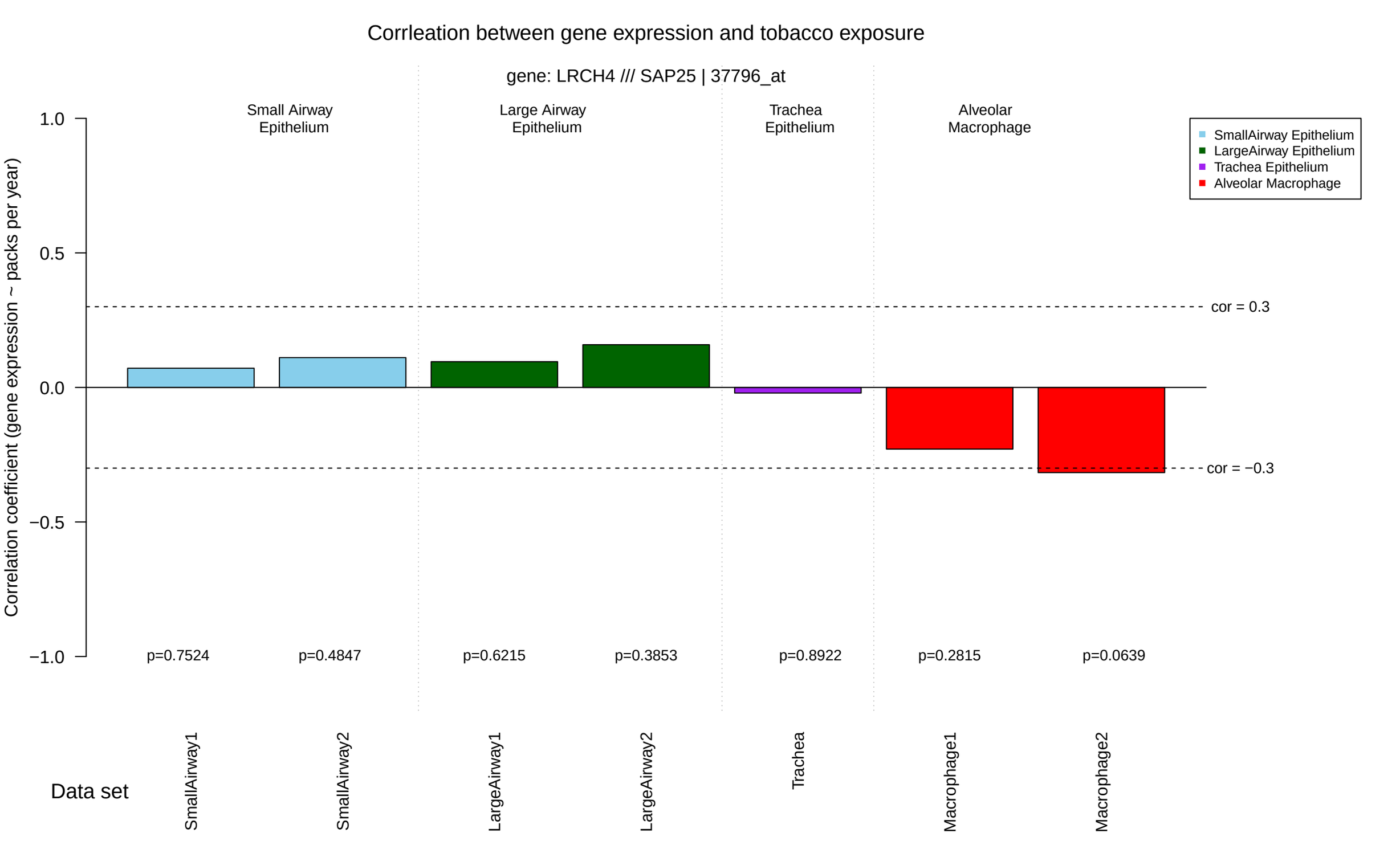

set: LargeAirway1 (p=0.2372)
LRCH4 /// SAP25 | 37796_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5214)
LRCH4 /// SAP25 | 37796_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3634)
LRCH4 /// SAP25 | 37796_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3707)
LRCH4 /// SAP25 | 37796_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.186)
LRCH4 /// SAP25 | 37796_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LRCH4 /// SAP25 | 37796_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7524
p=0.4847
p=0.6215
p=0.3853
p=0.8922
p=0.2815
p=0.0639
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2